

# ERYTRA EFLEXIS PREVENTIVE MAINTENANCE

Form version 1.0

40.000 tests  
6 Years

|                           |                |                          |     |
|---------------------------|----------------|--------------------------|-----|
| <b>Instrument:</b>        | Erytra Eflexis | <b>Form Version:</b>     | 1.0 |
| <b>Serial Number:</b>     |                | <b>Code:</b>             |     |
| <b>Installation Date:</b> |                | <b>Maintenance Date:</b> |     |

|                       |  |                       |  |
|-----------------------|--|-----------------------|--|
| <b>Customer Site:</b> |  | <b>Customer Name:</b> |  |
| <b>Department:</b>    |  | <b>E-Mail:</b>        |  |
| <b>Tel:</b>           |  |                       |  |

|                               |  |                |
|-------------------------------|--|----------------|
| <b>Service Engineer Name:</b> |  |                |
| <b>Department:</b>            |  |                |
| <b>Tel:</b>                   |  | <b>E-Mail:</b> |

|                                  |
|----------------------------------|
| Number of tests of last year PM: |
| Current number of tests:         |

| PARTS REPLACEMENT                  |                          |       | Instrument OFF |
|------------------------------------|--------------------------|-------|----------------|
| ACTION                             | Checked/Done             | Value | Comment        |
| Probe (PR)                         | <input type="checkbox"/> |       |                |
| Toothed belt T2.5/245 (PIP Y-axis) | <input type="checkbox"/> |       |                |
| Toothed belt T2.5/245 (PIP X-axis) | <input type="checkbox"/> |       |                |
| Toothed belt T2.5/245 (CAT Y-axis) | <input type="checkbox"/> |       |                |
| Toothed belt T2.5/245 (CAT X-axis) | <input type="checkbox"/> |       |                |
| Aspiration pump VPB                | <input type="checkbox"/> |       |                |

| VISUAL / PHYSICAL INSPECTIONS & CLEANING                                                                                                                                  |                          |       | Instrument OFF |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-------|----------------|
| ACTION                                                                                                                                                                    | Checked/Done             | Value | Comment        |
| <b>GENERAL</b>                                                                                                                                                            |                          |       |                |
| General appearance                                                                                                                                                        | <input type="checkbox"/> |       |                |
| Output cables and connectors                                                                                                                                              | <input type="checkbox"/> |       |                |
| Grounding continuity test. Switch off and unplug the electrical appliance. Check with a multimeter electric continuity between earth terminal of IEC inlet appliance and: |                          |       |                |
| Screw attaching centrifuge 1 to chassis                                                                                                                                   | <input type="checkbox"/> |       |                |
| Screw attaching centrifuge 2 to chassis                                                                                                                                   | <input type="checkbox"/> |       |                |
| <b>MONITOR ARM</b>                                                                                                                                                        |                          |       |                |
| Check that the arm movement is smooth                                                                                                                                     | <input type="checkbox"/> |       |                |
| Check that the arm blocking screw prevents undesired impacts of the monitor against the instrument chassis                                                                | <input type="checkbox"/> |       |                |
| <b>DOOR</b>                                                                                                                                                               |                          |       |                |
| Check that movement is smooth                                                                                                                                             | <input type="checkbox"/> |       |                |
| Check that door does not hit during automatic opening                                                                                                                     | <input type="checkbox"/> |       |                |
| Check that the latch enters smoothly when closing                                                                                                                         | <input type="checkbox"/> |       |                |
| <b>OTHER</b>                                                                                                                                                              |                          |       |                |
| If UPS is connected, check battery level                                                                                                                                  | <input type="checkbox"/> |       |                |
| <b>CARDS MODULE GATE</b>                                                                                                                                                  |                          |       |                |
| Check that movement is smooth                                                                                                                                             | <input type="checkbox"/> |       |                |
| <b>SAMPLES AND REAGENTS ENTRY (SRE)</b>                                                                                                                                   |                          |       |                |
| Check sample tube holders                                                                                                                                                 | <input type="checkbox"/> |       |                |
| Check reagent vial/bottle lodgings                                                                                                                                        | <input type="checkbox"/> |       |                |
| <b>CARD DRAWERS</b>                                                                                                                                                       |                          |       |                |
| Check card racks centerers and retainers                                                                                                                                  | <input type="checkbox"/> |       |                |
| <b>DILUTION CUP</b>                                                                                                                                                       |                          |       |                |
| Check dilution cup drain tubing                                                                                                                                           | <input type="checkbox"/> |       |                |
| <b>CENTRIFUGE MODULE</b>                                                                                                                                                  |                          |       |                |
| <a href="#">Removing centrifuge pot</a>                                                                                                                                   |                          |       |                |

|                                                                                           |                          |
|-------------------------------------------------------------------------------------------|--------------------------|
| Clean* pot of the centrifuge                                                              | <input type="checkbox"/> |
| Check rotor and card holders status                                                       | <input type="checkbox"/> |
| Check that rotor movement is smooth                                                       | <input type="checkbox"/> |
| <b>PIP MODULE</b>                                                                         |                          |
| Check belts tension and status                                                            | <input type="checkbox"/> |
| Check chains, wiring and tubing status                                                    | <input type="checkbox"/> |
| Check/clean* barcode reader mirrors                                                       | <input type="checkbox"/> |
| <b>CAT MODULE</b>                                                                         |                          |
| Check belts tension and status                                                            | <input type="checkbox"/> |
| Check chains and wiring status                                                            | <input type="checkbox"/> |
| Check gripper pins, centerer and fingers movement/status                                  | <input type="checkbox"/> |
| Check/clean* barcode reader mirror                                                        | <input type="checkbox"/> |
| <a href="#">Check/clean* presence detector</a>                                            | <input type="checkbox"/> |
| <b>FLUIDICS MODULE</b>                                                                    |                          |
| Check status of the fluidic input/output system for each bottle pos.                      | <input type="checkbox"/> |
| Check status of the bottles                                                               | <input type="checkbox"/> |
| Perform a fluidic circuit general cleaning*                                               | <input type="checkbox"/> |
| Check tubing                                                                              | <input type="checkbox"/> |
| Check the solution lines filters status                                                   | <input type="checkbox"/> |
| Check vacuum filters status                                                               | <input type="checkbox"/> |
| Check that the main fluidic manifold has no scratches nor leakages                        | <input type="checkbox"/> |
| <b>READER MODULE</b>                                                                      |                          |
| <a href="#">Removing reader cover</a>                                                     |                          |
| Check/clean* reader pattern                                                               | <input type="checkbox"/> |
| Check/clean* reader camera optics                                                         | <input type="checkbox"/> |
| Check reader foam status                                                                  | <input type="checkbox"/> |
| <b>*If cleaning is required, please note it in the comments box.</b>                      |                          |
| <b>LUBRICATIONS</b>                                                                       |                          |
| <i>Instrument OFF</i>                                                                     |                          |
| <b>ACTION</b>                                                                             | <b>Checked/Done</b>      |
| <b>DOOR</b>                                                                               |                          |
| <a href="#">Lubricating general module</a>                                                |                          |
| Lubricate door spring and door guides (grease)                                            | <input type="checkbox"/> |
| <b>CARD DRAWERS</b>                                                                       |                          |
| <a href="#">Lubricating general module</a>                                                |                          |
| Lubricate drawers latch (grease)                                                          | <input type="checkbox"/> |
| Lubricate linear guides (grease)                                                          | <input type="checkbox"/> |
| <b>PIP MODULE</b>                                                                         |                          |
| <a href="#">Lubricating pipetting arm module</a>                                          |                          |
| Lubricate X-axis linear guide (oil)                                                       | <input type="checkbox"/> |
| Lubricate Y-axis linear guide (oil)                                                       | <input type="checkbox"/> |
| Lubricate horns linear guide (oil)                                                        | <input type="checkbox"/> |
| Lubricate probe leadscrew (grease)                                                        | <input type="checkbox"/> |
| Lubricate probe washer holder cam (grease)                                                | <input type="checkbox"/> |
| Lubricate horns rack and pinion system (grease)                                           | <input type="checkbox"/> |
| <b>CAT MODULE</b>                                                                         |                          |
| <a href="#">Lubricating card transport arm module</a>                                     |                          |
| Lubricate X-axis linear guide (oil)                                                       | <input type="checkbox"/> |
| Lubricate Y-axis linear guide (oil)                                                       | <input type="checkbox"/> |
| Lubricate gripper linear guide (oil)                                                      | <input type="checkbox"/> |
| Lubricate gripper leadscrew (grease)                                                      | <input type="checkbox"/> |
| Lubricate gripper pins (oil)                                                              | <input type="checkbox"/> |
| Lubricate gripper centering system                                                        | <input type="checkbox"/> |
| <b>FLUIDICS MODULE</b>                                                                    |                          |
| <a href="#">Lubricating fluidics module</a>                                               |                          |
| Lubricate bottle pin entry for each bottle position (grease)                              | <input type="checkbox"/> |
| <b>GENERAL</b>                                                                            |                          |
| Check the general appearance of the instrument and if necessary clean the affected parts. | <input type="checkbox"/> |
| <b>FUNCTIONAL CHECKINGS &amp; ADJUSTMENTS</b>                                             |                          |
| <i>Instrument ON</i>                                                                      |                          |
| <b>ACTION</b>                                                                             | <b>Checked/Done</b>      |
| <b>DILUTION CUP</b>                                                                       |                          |

[Check proper stirrer operation  
\(PROFILE 0: 400 ± 40 rpm\)](#) ☐

#### CENTRIFUGE MODULE

☐

[Centrifugation verification \(PROFILE 0:  
1206 rpm ± 36 rpm with duration  
of 550 s ± 10 s\)](#)

#### PIP MODULE

[Check presence detector alignment](#) ☐

[Verify the CPD sensor with Leak-  
tightness test in TSS](#) ☐

[Check probe offset in TSS](#) ☐

#### CAT MODULE

[Check gripper-presence detector  
alignment](#) ☐

#### READER MODULE

[Removing reader cover](#)

[Perform reader adjustment](#) ☐

[Perform gripper adjustment in card  
reader area](#) ☐

#### SOFTWARE

Folders inspection ☐

#### COMPUTER AND TOUCHSCREEN

SSD available space ☐

Touchscreen operability (touch  
functionality in all screen areas) ☐

#### QUALIFICATION

| ACTION                                                                    | Checked/Done             | Value | Comment |
|---------------------------------------------------------------------------|--------------------------|-------|---------|
| <a href="#">Perform Operational Qualification (DG-<br/>AT-TSM-000150)</a> | <input type="checkbox"/> |       |         |